Mastara (ማስታራ) — Software Requirements Specification v3.1

Rebuild baseline · Functional requirements by partition · UI flow design · Question Bank first

Field	Value
Product	Mastara (ማስታራ) — online-first learning app for Ethiopian university freshmen, with optional downloads for offline study
Document ID	MASTARA-SRS-001
Version	3.1 — admin screens aligned to the approved design; supersedes v2.0, v2.1, v2.2 and v3.0
Status	Draft for product-owner approval
Date	15 October 2026
Student client	React Native with Expo (Android first, iOS supported)
Admin platform	Web portal built with Next.js
Backend	NestJS
Database	**PostgreSQL** (replaces MySQL)
Package manager	**npm** with workspaces (replaces pnpm)
Roles	STUDENT, ADMIN (Admin portal holds every operational asset)
Currency	ETB
App size target	**Under 20 MB** (hard release gate)
Normative language	MUST = mandatory · SHOULD = recommended · MAY = optional

1. Purpose and how to read this document

This is the single authoritative specification for Mastara. It rewrites v2.0 using every instruction in the product brief from 6 to 15 October 2026. **Where two instructions conflict, the later one wins.** Section 2 lists each conflict and how it was resolved, so the product owner can overrule any reading.

Functional requirements are grouped into partitions (Section 8). Each partition has a definition, a workflow, a UI flow and testable requirements. IDs use a partition letter (for example FR-E12). Priority: **P0** launch-blocking, **P1** launch-expected, **P2** later.

2. Authoritative decisions and conflict resolution

#	Decision (v3.0)	Replaces in v2.0
D1	Stack: Expo React Native, Next.js admin, NestJS API, PostgreSQL, npm workspaces monorepo	MySQL 8, pnpm
D2	Release package MUST be under 20 MB, measured as the arm64 download size from the Android App Bundle	50 MB target
D3	Only two roles: STUDENT and ADMIN. Admin owns the assets of every other role (editor, reviewer, finance, support)	unchanged
D4	**Local-first account.** A student creates an account on the phone, offline if needed, and uses free content at once. Nothing is written to the remote database until the first purchase. Phone number registration, remote account creation and device binding happen at purchase	Register, then Device Registration at launch
D5	**Online-first content.** Every content item is fetched online the first time. The student may download an item to the app, use it offline, and delete it later to free space	Offline-first with mandatory caching

#	Decision (v3.0)	Replaces in v2.0
D6	**Continue Learning is removed** from Home and from the specification	Home section and FR-044 resume
D7	Home is **card-based**. COC Exam card uses the same card style as course cards	List-style Home
D8	Material tabs are Modules, Notes, Amharic Note, with Amharic Note next to Notes. **No Amharic/English toggle exists inside any content page.** Amharic Notes are their own content items, not a second language view of one record	One bilingual record and in-page tab switch (D8, FR-023)
D9	**Content protection:** screenshots and screen recording are blocked on every content screen (modules, notes, Amharic notes, tests, exams, COC). See 11.4 for what this can and cannot stop	Partial protection
D10	Light and dark theme toggle on every screen; brand colours unchanged	Light only
D11	**Scroll-first reading.** Content pages, quizzes and exams scroll continuously. A Next button sits at the bottom for students who prefer to tap	Next-driven pages
D12	After any exam or COC exam, a **Review Answer** button lists every taken question with the correct answer and a description. Descriptions have **Show / Hide** buttons	Result only
D13	Exam duration is set by Admin **on the exam set**. The Time Allowed input is removed from question forms	Duration on exam
D14	Question forms and list screens follow Section 7.4 (latest field list)	Earlier forms
D15	COC Exam is a standalone Home module, not a tab of Exams	unchanged

#	Decision (v3.0)	Replaces in v2.0
D16	GPA is out of scope everywhere	unchanged
D17	Payment is a manual receipt screenshot (CBE Mobile Banking, CBE Birr, Telebirr) verified by Admin	unchanged
D18	Each course carries a **free-trial item count** set by Admin when the course is created	Flag-only trial
D19	**Build order:** review both codebases, then deliver the Question Bank module first, in small sub-phases	Roadmap order

2.1 Conflicts found and how they were resolved

Conflict	Resolution used here	Confirm?
6 Oct asks for device registration right after login; 15 Oct defers phone registration to purchase	Device binding starts at first purchase. Before it, the app is local-only and free content is not bound to a device	Q1
11 Oct says exam time is set by Admin; 15 Oct removes the Time Allowed input	Duration lives on the exam set (Mid, Final, Mock, COC set) and is not a field of the question form	Q2
11 Oct 8 PM removes Quiz 1, 2, 3 tabs; 12 Oct lists Quiz 1–3 as sets and scrolling scope	Quiz tabs are removed from the **Admin Questions** module. Quizzes are authored in Admin **Content → Tests** (Quiz 1, 2, 3) and students keep Quiz 1–3 in the Tests tab	Q3
COC form has no Course field, but v2.0 promised a per-course breakdown	The design shows Course on the COC form as optional. COC questions may carry a course, and the per-course breakdown covers tagged questions	Q4

Conflict	Resolution used here	Confirm?
Access/Permission removed from question forms	Questions carry no permission of their own. Access is set on the exam set (Access / Permission, default Paid) and by course entitlement	—
Topic removed from question forms, but kept for content	Topic stays on Module, Note, Amharic Note and Quiz content. It is removed only from the Questions module	—
15 Oct says Exam Set Id removed; 12 Oct had `examSetId` in the COC schema	Removed from question forms and lists. The Build set dialog keeps an Exam Set ID label for the set	—
20 MB target with Expo, Hermes and encrypted SQLite	Kept as a hard gate. It is tight (see risk R1 in Section 14)	Q5

3. Product definition, scope and roles

3.1 Definition

Mastara helps freshmen find materials, understand them, practise, and sit exams, even on cheap phones with unreliable internet. A student sets University, Program, Semester and Stream, adds courses to My Course List, tries free items, then buys access. Learning uses modules, notes, Amharic notes, quizzes, Mid, Final and Mock exams, and a standalone COC Exam.

3.2 In scope

- Local account creation, Profile Setup, My Course List, free trial
- Phone registration, one-device binding and release at first purchase
- Stream-filtered catalog (Natural, Social, Common)
- Course tabs Material, Tests, Exams; standalone COC Exam on Home
- Online delivery, optional download, delete downloads
- Answer review with Show / Hide descriptions
- Bundles, single courses, extra courses, Full Freshman package, receipt payment, Admin verification, automatic expiry

- Admin web portal: catalog, content, Scan to PDF, Questions, Products and Pricing, payments, entitlements, devices, notifications, support, audit, reports
- Screen-capture blocking, dark and light themes, colourful icon system

3.3 Out of scope

GPA calculation; Continue Learning; automated payment gateways; live remote proctoring; AI tutor and recommendations; roles other than Student and Admin; iOS store billing.

3.4 Roles and authorization

Role	Responsibility
STUDENT	Create local account, choose scope, use free items, register phone and buy access, learn, sit exams, review answers, download or delete content, upload payment proof, release own device
ADMIN	Everything else, including students, catalog, content, scanning, questions, products, prices, durations, payments, entitlements, devices, notifications, support, audit, reports

Server rule on every request: `ALLOW = authenticated + role permission + resource scope + resource state + business rule`. Paid content opens only when `STUDENT AND device_matches AND entitlement_active(course) AND status = PUBLISHED`. Free-trial items open under the rule in 4.6. The client never sends a role; a forged role is ignored and logged.

4. Commerce and access model

Every number is Admin-managed configuration. Defaults come from the brief.

4.1 Course classification

Every course has one stream: NATURAL, SOCIAL or COMMON. Common courses appear under both stream filters.

4.2 Products

Product	Default price	Rule
Semester Bundle	200 ETB	Student picks 7–10 courses of one semester and stream (range Admin-configurable). One flat price per semester
Extra Course	100 ETB first, then 50 ETB	Courses beyond the bundle maximum use the Admin tiered schedule
Single Course	Admin-defined	May be bought before any bundle
Full Freshman Package	350 ETB per stream	All stream courses plus Common courses, both semesters, with COC and exams
Both Streams	Admin-defined (default 2 × 350)	Natural, Social or both in one order

4.3 What paid access unlocks

A paid course unlocks all its protected assets: modules, notes, Amharic notes, PDFs, quizzes, Mid, Final and Mock exams, university material and COC access. A student who bought specific courses MUST NOT open unpaid courses. Unpaid courses stay visible as advertisement.

4.4 Duration and expiry

Admin sets `duration_days` per product. Approval creates entitlements with `starts_at` and `expires_at`. The server is authoritative: when `expires_at` passes, protected content locks and downloaded copies stop opening. Device clock changes MUST NOT extend access.

4.5 Pricing rules

ID	Rule
PR-1	A bundle with fewer than the minimum or more than the maximum courses is rejected on client and server
PR-2	Going beyond the maximum is possible only by adding Extra Courses, each priced in the summary
PR-3	Price is computed on the server. The client amount is display-only
PR-4	A Full package covers any single or bundle purchase of the same stream; no double charge for owned courses
PR-5	An actively entitled course cannot be bought again until the renewal window
PR-6	Price changes apply to new orders only
PR-7	Single purchases are not credited to a later bundle unless `credit_single_to_bundle` is on (default off)

4.6 Free trial

When Admin creates a course, Admin enters **Free trial items** (a number). The first N items, in Admin order, open without payment:

Area	Free-trial behaviour
Modules, Notes, Amharic Notes	First N items open fully; the rest show a lock
Quizzes	First N quiz chapters open with full feedback
Exams	Locked, unless Admin marks one sample paper per category as free
COC Exam	A sample set with an Admin-set question count

Area	Free-trial behaviour
Expiry	None by default; Admin may set trial expiry days per course
After purchase	Trial progress and answers carry over

A trial never creates an entitlement. Because free students have no remote account (D4), free items are served as public, rate-limited, watermarked assets keyed to an anonymous install ID. Admin can still flag individual items free or locked to override the count.

4.7 Purchase flow

Store → stream → product → course selection (bundle 7–10) → price summary → **phone registration (first purchase only)** → device binding → payment instructions → pay → upload receipt → reference → submit → Admin queue → approve → entitlements active → expiry → lock → renew.

5. Information architecture and navigation

5.1 Hierarchy

University → Program → Semester → Course (stream) → Material | Tests | Exams. COC Exam sits beside this tree, reached from Home.

5.2 Student navigation

Level	Structure
Bottom navigation	Home · Course · Progress · Profile
Home (cards)	Courses · COC Exam · Learning Status · Payment Alerts · Store Status
Course screen	Search field on top; filter tabs **All · Ongoing · Completed**; My Course List as cards
Course top tabs	Material · Tests · Exams
Material lower tabs	Modules · Notes · Amharic Note
Tests lower tabs	Quiz 1 · Quiz 2 · Quiz 3, each listing its chapters

Level	Structure
Exams lower tabs	Mid · Final · Mock, each listing papers by university
COC Exam	Standalone module opened only from the Home COC card
Profile	Personal info · University, Program, Semester, Stream · My Courses · Subscription and Store · Downloads and Storage · Security and Device · Settings (language, theme) · Support

Store, Notifications and Support open from Home cards, lock prompts and Profile.

5.3 Admin navigation

The sidebar follows the approved design: Dashboard · Students · Courses · Content (Material: Modules, Notes, Amharic Notes; Tests: Quiz 1, Quiz 2, Quiz 3) · Scan to PDF · **Questions** (Mid Exams, Final Exams, Mock Exams, COC Exams) · Payments · Pricing · Support · Announcements · Reports · Audit Log. Universities, Programs and Semesters are tabs of Courses. Devices and Entitlements are tabs of Students. Payment Queue sits under Payments, tickets under Support. Exam sets open from each Questions child.

5.4 Access states on every card

State	Presentation	Primary action
Available	Normal card	Open
Free trial	Free badge	Open
Locked	Lock icon and reason	Buy or Renew
Pending	Verification indicator	View status
Expiring	Warning with days left	Renew
Expired	Locked, expiry date	Renew
Downloaded	Download badge	Open, or Delete download
Offline	Offline badge	Use downloaded items

6. Student UI flow design

6.1 Primary workflow

Install → Splash → Create account (local, works offline) → Profile Setup → Home. A returning student with a complete profile goes straight to Home. Device registration is not part of launch; it occurs inside the purchase flow (6.8).

Step	System behaviour	On failure
Install	Package under 20 MB; no textbooks, PDFs or large images bundled	—
Splash	Load config, local DB, theme, language, session	Offline splash opens the local profile
Create account	Name, password or PIN and optional email stored locally; no server call	Field errors
Profile Setup	University, program, semester, stream, then courses (6.7). Save is disabled until the four scope fields are set	Retry saved locally
Home	Load cards for the saved profile; fetch content metadata when online	Cached Home with offline badge

6.2 Home layout (card-based)

Card	Function	Icon accent
Courses	Opens the Course screen with My Course List	Blue
COC Exam	Opens the standalone COC module; same card size and style as a course card	Orange
Learning Status	Progress, completion, download and sync status	Green
Payment Alerts	Pending, rejected, expiring, expired	Red/amber
Store Status	Purchased, pending, expired, promotional	Purple

There is no Continue Learning card. Each card has a coloured icon tile, a title, one line of status and a chevron.

6.3 Course screen

Search field on top, then three filter tabs. **All** shows every course in My Course List; **Ongoing** shows courses with progress above 0% and below 100%; **Completed** shows courses at 100%. Search matches title and code inside the active tab. Each card shows the course icon, name, stream chip, progress bar and an access state. The overflow menu offers Edit course list, Download manager and Help.

6.4 Reading flow (Modules, Notes, Amharic Note)

Topic or chapter → check content lock → read.

1. Content opens as one **continuously scrolling page**. The student reads by scrolling up and down.
2. A **Next** button sits under the page for tap navigation to the next topic.
3. No language toggle appears inside the content. Amharic Note is a separate item in its own tab beside Notes.
4. Mixed Amharic, English and formulas render correctly (Noto Sans Ethiopic and KaTeX, see NFR-07).
5. Screen capture is blocked on this screen (D9).
6. First open fetches online. A **Download** action stores the chapter on the phone; **Delete download** removes it.

6.5 Quiz flow

Topic → lock → practice → question → answer → immediate feedback → next.

1. Validate access. Load the question set online, or from a download.
2. Questions appear in one scrolling list or one-by-one; both end with a bottom Next button.
3. The answer is written locally at once and feedback with the explanation appears with **Show / Hide**.
4. Completion shows score and **Review Answer**.
5. Quiz 1, 2 and 3 have independent sets, settings and progress.

6.6 Mid, Final, Mock and COC exam flow

Click → check content lock → rules screen (duration and question count, both from the exam set) → start → **scrollable All Questions** → answer → submit → result → **Review Answer**.

1. Questions are listed in one scrollable page. A numbered palette jumps to any question. A bottom Next button supports tap navigation.
2. Timer runs on monotonic elapsed time and auto-submits at zero.
3. Submit asks for confirmation. Result visibility follows Admin policy.
4. **Review Answer** shows every taken question with the student's answer, the correct answer and the answer description. Each description has a **Show** and **Hide** button (hidden by default).
5. COC Exam has its own screens, a mixed-course pool, practice mode (untimed, feedback) and exam mode (timed). Unpaid students see a sample set.
6. Exam state machine: READY → STARTED → IN_PROGRESS → SUBMITTING → SUBMITTED → RESULT_AVAILABLE, with IN_PROGRESS → LOCAL_RECOVERY → IN_PROGRESS on interruption.

6.7 Profile Setup, Profile and My Course List

First run: Create account → Profile (setup mode) → University → Program → Semester → Stream → select courses → Save → Home. Later: Profile → edit fields or My Courses → add or remove → Save → Course screen updates.

Rule	Behaviour
Required	University, program, semester, stream. Courses are optional
Cascade	Programs follow the university, semesters follow the program, courses follow semester and stream, with Common always shown
Add	Any published course in scope, paid or not. Adding never grants access
Remove	Confirmation; progress is kept; a paid course warns that removal does not cancel the entitlement

Rule	Behaviour
Scope change	Confirmation; courses outside the new scope leave the list and their progress is archived
Offline	All edits work offline and stay local until the first purchase, then sync

6.8 Store, registration and payment flow

Store → stream filter → product → course selection (preselected from My Course List) → price summary → **Register phone** (first purchase only) → bind device → payment instructions → import screenshot → compress under 300 KB → transaction reference → submit → pending → result.

Screen	Content
Phone registration	Phone number and verification, then the local profile and progress upload to the server. Shown once
Payment instructions	Tabs for CBE Mobile Banking, CBE Birr, Telebirr; Admin account number with Copy; exact amount
Proof upload	Gallery picker, preview, auto-compression under 300 KB, reference field (FT… or Telebirr transaction ID)
Status	Pending, Approved, Rejected with reason and Resubmit

6.9 Device lock and release

Scenario	UI
First purchase	Silent bind after phone registration; confirmation toast

Scenario	UI
Second phone	Full-screen modal, no content reachable: "ይህ አካውንት በሌላ ስልክ ላይ ተመዝግቧል። ስልክ ለመቀየር እባክዎ ዋናውን ስልክ ያጥፉ/ይውጡ ወይም አድሚኑን ያነጋግሩ" with Contact Admin and Log Out
Release	Profile → Security → Release Device & Log Out → confirm → binding revoked, local credentials and downloads cleared
Admin recovery	Admin revokes from Devices with reason; audited

6.10 Downloads and storage

Download manager lists downloaded items with size, per-item **Delete**, and **Delete all**. Downloads are encrypted and tied to the account and device once bound. Expiry, release and revocation make downloaded paid items unreadable. Free-trial items can be downloaded and removed if Admin withdraws them. Low storage warns before a download.

6.11 Visual system

- **Brand and colours are unchanged** from the earlier design: light blue-white background, white rounded cards (12–16 dp), primary blue for active states, Noto Sans Ethiopic for Amharic.
- **Colourful icons everywhere.** Each feature has its own accent: Courses blue, Material sky, Tests pink, Exams amber, COC orange, Learning Status green, Payment Alerts red, Store purple, Profile teal. Each course gets its own icon and colour, set by Admin or assigned from the palette.
- **Theme:** a Light / Dark toggle in Settings and in the top bar menu, applied to every screen, remembered locally. Dark mode uses a dark navy surface, with accents lightened to keep contrast of at least 4.5:1.
- Tap targets at least 48 dp; modern card layout, soft shadows, short motion under 250 ms.

7. Admin web portal UI flow design (Next.js)

Every admin screen now follows one design, taken from the Exam Management, Test Management, Content, Courses and Scan to PDF screens in the approved mockups. The Content, Quizzes and Scan to PDF screens used other layouts and are refactored to match. **Every list tab has pagination, and every row has Edit, Delete and an Active / Inactive toggle.** Delete asks for confirmation; Inactive hides the item from students without removing it. Every sensitive action writes an audit record.

7.0 Shared admin shell and design rules

The shell comes from the Exam Management screens. Content, Quizzes and Scan to PDF currently use a lighter "Admin Portal" sidebar or a plain white top bar and MUST be moved onto it.

Element	Rule
Sidebar	Dark navy, Mastara logo with an "Admin" label, one icon per item, expandable groups, active item filled blue, active child shown with a dot. A count badge on Payments shows pending receipts
Top bar	Global search (placeholder follows the page, for example "Search questions & COC..." or "Search scan to PDF..."), Admin menu with theme toggle, Log out button
Page header	Back link inside a workflow (for example "Back to exam sets"), page title, a draft badge such as "Draft v3" where versions exist, and one **Preview as student** button at top right on every authoring page
Tabs	Pill tabs, active tab filled blue. Tab order always equals the sidebar order
Cards	White rounded cards on a light blue-white page, with a card heading such as "Create Mid Exam Question"

Element	Rule
Forms	Label above each field, "Select a ..." placeholders, 5-column then 3-column grids, character counters on long text (Question 0/1000, Answer Description 0/500)
Formula hint	One hint line under every question and description field: "You can use mathematical formulas", with examples such as `$x^2$`, a fraction and a square root, and a live preview
Action bar	Bottom of the card. Left: **Add Question** (or **Add New Chapter**). Right: **Reset**, **Save Draft**, **Publish** (Publish filled blue). The same bar is used on question, quiz and content forms
Lists	Filters row, table, pagination, then Edit, Delete and the Active toggle on each row

Differences found in the mockups and how this SRS settles them:

In the mockups	Settled rule
Questions children appear as Final, Mid, Mock, COC in one screen and Mid, Final, Mock, COC in others; tabs read Mid, Final, COC, Mock	Order is **Mid, Final, Mock, COC** in the sidebar and in the tabs
Sidebar item reads "Questions & COC" on older screens	Renamed **Questions** with four children
Content editor is titled "Courses" and its chapter list is headed "Courses"	Page title is the content type (Modules, Notes, Amharic Notes, Test Management); the list is headed **Chapters**
Content tabs read Notes, Modules, Amharic Notes	**Modules, Notes, Amharic Notes**, with Amharic Notes beside Notes
Helper text says English and Amharic are fields of one note	**Removed.** Amharic Notes are their own items (D8)

In the mockups	Settled rule
"Preview as student" on content, "Review as student" on exams	One label: **Preview as student** (Review Answer is only the student button after an exam)
Quiz form says Answers and Description (Explanation); exam form says Options and Answer Description	Both use **Options**, **Correct Answer** and **Answer Description**
Top-of-page Save draft and Publish on Content and Quizzes; bottom bar on exams	Bottom action bar everywhere
Scan to PDF details include a Language field	**Removed** (D8). The PDF carries no language field
Scan to PDF offers "Use 8 sample pages"	Development helper only; hidden in production

7.1 Courses page

Courses has tabs **Universities, Programs, Semesters, Courses**, each with Add, search, pagination, Edit, Delete and the Active toggle. Workflow: Courses → Add course → fill fields → Save → row appears in the list.

Field	Rule
University	Dropdown with an **All** option. All makes the course available to every university
University logo	Shown beside the dropdown and on the course row once a university is chosen
Program, Semester, Stream	Cascading selectors; Stream is Natural, Social or Common
Course code and title	Required
Icon and colour	Chosen from the palette or auto-assigned
Free trial items	Number of items open without payment for this course (D18)
Trial expiry days	Optional

Field	Rule
Promotional flag	Shows the course as advertisement
Status	Active / Inactive toggle

7.2 Content page (Materials and Tests)

Sidebar: **Content** expands to **Material** (Modules, Notes, Amharic Notes) and **Tests** (Quiz 1, Quiz 2, Quiz 3). Materials and Tests share one two-pane layout.

Pane	Content
Scope row (above the panes)	University, Program / Department, Semester, Stream, set once and kept while editing
Left pane	Course dropdown, a **Chapters** list (Chapter 1, Chapter 2, …) with the selected chapter highlighted, and an **Add New Chapter** button. Modules and topics are added as chapters
Right pane header	Tabs for the content type (Modules, Notes, Amharic Notes; or Quiz 1, Quiz 2, Quiz 3) and a **Free trial** checkbox that marks the chapter free regardless of the course count
Action bar	Add New Chapter / Add Question on the left; Reset, Save Draft, Publish on the right

Materials screen (Modules, Notes, Amharic Notes): fields are Chapter title, Topic, Chapter content (rich text, `$...$` formulas with offline preview) and Description. Buttons **Attach PDF** and **Open scan to PDF** sit under the content. There is no language option anywhere on the page.

Tests screen (Test Management): opening Quiz 1, 2 or 3 shows a list first:

Part	Rule
Header	Title "Test Management", line "Review, filter and edit the questions of a quiz", and an **Add Question** button

Part	Rule
Tabs	Quiz 1, Quiz 2, Quiz 3 (these mirror the sidebar children)
Filters	Course, Type, Status, Chapter
Columns	ID, Question, Course, Type, Chapter, Status, Edit (plus Delete and Active toggle)
Pagination	Required

Add Question / Edit opens the quiz question form on the same shell: Chapter title, Question type (Multiple Choice, True / False, Short Answer, Blank Space, each with its own dedicated inputs), Question number (within the chapter), Question, Options with a radio mark for the correct one and Add Option, Correct Answer, Answer Description (typed, or attached as image or PDF), then the action bar. The Quiz 1–3 tabs are removed only from the Questions module, not from here.

7.3 Scan to PDF

Scan to PDF is a four-step wizard shown as four pill steps: **1 Import → 2 Order and edit → 3 Convert → 4 Publish**. It serves COC, Exams, Materials and Tests.

Step	Screen	Rules
1 Import	Dashed drop zone titled "Add the scanned pages" with the line "6 or more images are fully supported. Up to 200 per PDF." Buttons **Choose images** (also camera capture) and a counter "N images added"	**Next: order and edit** stays disabled until at least one image is added
2 Order and edit	Left card "N images, drag to reorder" with numbered thumbnails and **Rotate, Crop, Delete** buttons; right card **PDF details**; footer "Pages: N · Order as shown" and **Convert to one PDF**	Edits are non-destructive until convert; page order is exactly the order shown

Step	Screen	Rules
3 Convert	"PDF ready" with chips for page count and size in KB, **Preview PDF**, **Back to edit**, **Continue to publish**	Blank-page warning; a failure is retryable and keeps the source images
4 Publish	Summary rows Title, Access, Pages, Size; note "Published PDFs inherit entitlement protection"; buttons **Back, Save draft, Publish**	Title is filled automatically as "Course · Category" and can be edited

PDF details are driven by Category. Category is the first field; choosing it sets the rest. Access is always the last field (Paid course only, or Free).

Category	Fields
Modules, Notes, Amharic Notes	University → Program / Department → Semester → Stream → Course → Topic → Title → Description
Quiz 1, Quiz 2, Quiz 3	Same as above, plus an Answer typed on the keyboard or attached as image or PDF
Mid Exam, Final Exam	University → Program / Department → Semester → Stream → Course → Year; answers by scanned image, PDF or typed in the form
Mock Exam	Program / Department → Semester → Stream → Course; same answer options
COC Exam	University → Program / Department → Year; each question's answer by image, PDF or typed text

7.4 Questions module (Exam Management)

Sidebar: Questions nests **Mid Exams, Final Exams, Mock Exams, COC Exams**. Each child opens on its **exam sets list**. Flow: Questions → a child → exam sets list → **Build set** (7.5) → set saved as draft → Exam Management page → **Create … Question** form → Add Question → Publish. The link **Back to exam sets** returns to the list.

Exam Management page: title "Exam Management", pill tabs **Mid Exams, Final Exams, Mock Exams, COC Exams**, then a card headed "Create Mid Exam Question" (or Final, Mock, COC). The same layout is used for all four kinds, including COC and Mock, which

had no tab bar in some mockups.

Question form (field layout as designed):

Kind	Row 1	Row 2
Mid and Final	University, Program / Department, Semester, Stream, Year	Course, Question Type, Question ID
COC	University, Program / Department, Semester, Stream, Year	Course (optional), Question Type, Question ID
Mock	Program / Department, Semester, Stream	Course, Question Type, Question ID

Then, for every kind: **Question** (0/1000, formula hint), **Options** (four by default, each with a radio mark and a delete icon, plus **Add Option**), **Correct Answer** (dropdown of the options), **Answer Description** (0/500, helper "This explanation will be shown to students after they answer the question"), and the action bar **Add Question, Reset, Save Draft, Publish**.

- Scope fields are filled from the exam set and may be changed per question.
- **Question ID** is a typed label (for example Q-1006). The field suggests the next free ID, which must be unique. **Question Number** is the auto-incremented identity that the list shows first.
- Question Type options: Multiple Choice, True / False (two fixed options), Short Answer (accepted answers), Blank Space (blank markers). Each shows its own inputs in place of the Options block.
- **Add Question** saves the current question as a draft in the set and clears Question, Options, Correct Answer and Answer Description, keeping the scope fields.
- **Publish** requires complete scope, a non-empty question, a correct answer and at least two options. Answer Description is recommended, not required.
- The students see the description in Review Answer behind Show / Hide.
- **Removed from every question form:** Set Exam Id, Topic, Permission / Access, Time Allowed. Access and time now live on the exam set (7.5).

List screen (inside each sub-category):

Column order	Notes
Question Number	Auto-incremented, shown first, before University
University	Not shown for Mock
Program, Semester, Stream	
Course	Blank for COC questions with no course
Year	Not shown for Mock
Question type, Question	
Status	Renamed from "Group Status". Draft, Published, Inactive

Filters for **University, Course, Year, Stream and Program** sit beside the Status filter. Exam Set Id and Grouped Question Ids are not columns. Pagination, Edit, Delete and the Active toggle apply to every row.

7.5 Exam sets and the Build set dialog

Each Questions child lists its exam sets with a **Build set** button. The dialog is titled "Build Mid Exam Set" (or Final, Mock, COC), with the line "Name the set and say who it is for. It is saved as a draft, and you add its questions next." Buttons: **Cancel** and **Build set**.

Kind	Fields in order
Mid and Final	Exam Set ID, University, Program / Department, Semester, Stream, Course, Access / Permission, Year, Allowed Time (min)
COC	Exam Set ID, University, Program / Department, Semester, Stream, Access / Permission, Year, Allowed Time (min)
Mock	Exam Set ID, Program / Department, Semester, Stream, Course, Access / Permission, Allowed Time (min)

- **Exam Set ID** is a typed label, unique within its kind (for example AASTU-FINAL-2026). It appears only in this dialog and on the sets list, never on question forms or question lists.
- **Access / Permission** is Paid by default; Free marks a sample paper that unpaid students can open.
- **Year** is an academic year such as 2026 / 2027.
- **Allowed Time (min)** defaults to 60 and is the exam duration students see (D13). This is where Admin specifies exam time.
- Quiz sets have no duration; their chapters are assigned in Content → Tests.
- Other set options (shuffle, feedback mode, result visibility) sit on the set's detail page.

7.6 Payment verification

Dashboard → Payments → Payment Queue → open receipt → check student, product, courses, amount and reference → zoom the image → Approve or Reject (reason mandatory) → confirm → entitlements created → notification → audit. The page shows the quoted price against the selected courses, a duplicate-reference warning and the student's history.

7.7 Products, pricing and duration

Pricing → select product → edit price, course range, extra-course tiers, duration → save → version recorded → applies to new orders.

7.8 Student, device and entitlement control

Students has tabs **Students, Devices, Entitlements**. Search a student → view profile, device binding, entitlements, payments → suspend, revoke device, grant or extend access (reason required) → audit. Admin may also correct a student's scope and course list.

8. Functional requirements by partition

Each partition lists its definition, workflow and requirements. Every requirement is testable and traceable (Section 15).

8.A Partition A — Identity, local account and device

Definition: who the student is, and which single phone may use a paid account. **Workflow:** create local account → use free content → first purchase → register phone → bind device → session → (release) → rebind.

ID	P	Requirement	Acceptance
FR-A01	P0	Create a local account (name, password or PIN, optional email) with no network	Works in airplane mode; stored encrypted on the phone
FR-A02	P0	Free content opens with the local account	No server account is needed
FR-A03	P0	Phone registration is requested only when the student tries to buy	Not shown at install, splash or profile setup
FR-A04	P0	On phone registration, create the remote account and upload the local profile and progress, idempotently	One student row per phone; repeat call changes nothing
FR-A05	P0	No remote record exists before the first purchase attempt	Database test: zero student rows for local-only users
FR-A06	P0	Login on the remote account with tokens	Valid login issues access and refresh tokens
FR-A07	P0	Secure hardware device ID (Keystore / Keychain) sent as `x-device-id`	Every protected request carries it
FR-A08	P0	One-device binding enforced on the server from first purchase	Second phone gets 403 `DEVICE_MISMATCH_LOCKED`
FR-A09	P0	Localized mismatch modal (Amharic and English)	Blocks all content; Contact Admin and Log Out
FR-A10	P0	Release Device & Log Out	Binding revoked; tokens, local credentials and downloads cleared
FR-A11	P0	Admin inspects and revokes bindings	Reason mandatory; audited

ID	P	Requirement	Acceptance
FR-A12	P1	Security event log	Actor, device hash, result, time
FR-A13	P1	Warn that uninstalling before the first purchase erases local progress	Notice shown in Profile and at account creation

8.B Partition B — Academic catalog and streams

Definition: the University → Program → Semester → Course tree and the stream filter.
Workflow: Admin builds the catalog → student picks scope → catalog filters by stream → card state comes from entitlement.

ID	P	Requirement	Acceptance
FR-B01	P0	Admin manages universities, each with a logo	Published universities appear in pickers
FR-B02	P0	Programs belong to a university; semesters to a program; courses to a semester	Correct cascading lists
FR-B03	P0	Course stream is NATURAL, SOCIAL or COMMON	Exactly one stream per course
FR-B04	P0	Stream filter in Course picker and Store	Natural shows Natural plus Common; Social shows Social plus Common
FR-B05	P0	University dropdown includes **All** when adding a course	An All course appears for every university
FR-B06	P0	University logo shown in the add-course form	Logo changes with selection
FR-B07	P0	**Free trial items** number saved per course	Server uses it to decide open items (FR-C17)
FR-B08	P0	Promotional flag per course	Visible as advertisement; protected content locked
FR-B09	P0	Pagination, Delete and Active / Inactive on every admin tab	Inactive items disappear from the student app

ID	P	Requirement	Acceptance
FR-B10	P1	Student may change semester scope	Progress retained per scope

8.C Partition C — Content, downloads and protection

Definition: everything a student reads. **Workflow:** Admin publishes → lock check → fetch online → read by scrolling → optional download → optional delete.

ID	P	Requirement	Acceptance
FR-C01	P0	Modules, Notes and Amharic Note tabs, with Amharic Note next to Notes	Tab order Modules, Notes, Amharic Note
FR-C02	P0	Amharic Notes are separate content items	No duplicate bilingual record is required
FR-C03	P0	No Amharic / English toggle inside any content page	UI audit finds none
FR-C04	P0	Content pages scroll continuously with a bottom Next button	Applies to modules, notes, quizzes and exams
FR-C05	P0	Content lock check before every protected open	Missing, pending or expired access is locked on server and client
FR-C06	P0	First access fetches content online	No content is bundled in the app
FR-C07	P0	Per-chapter download to encrypted app storage	Downloaded item opens offline
FR-C08	P0	Delete one download or all downloads	Space is freed; item re-fetches online
FR-C09	P0	Expiry, release or revocation makes downloaded paid items unreadable	Tested with a past `expires_at`
FR-C10	P0	**Block screenshots and screen recording** on every content screen	Android `FLAG_SECURE`; iOS capture detection hides content

ID	P	Requirement	Acceptance
FR-C11	P1	Dynamic watermark (phone or install ID) over content	Visible in any photo of the screen
FR-C12	P0	Offline LaTeX and Ethiopic rendering	No network request for fonts
FR-C13	P0	Content versioning; published versions are immutable	Traceable
FR-C14	P0	PDFs open only for entitled or free-trial students	Short-lived signed URLs
FR-C15	P1	Resumable chapter downloads, low-storage warning	Verified on a throttled network
FR-C16	P2	Reading position kept locally inside a topic	Not shown on Home
FR-C17	P0	Server opens the first N items of a course free, N from FR-B07, plus Admin overrides	A forged request for a locked item returns 403

8.D Partition D — Quizzes, exams, COC and review

Definition: all question-driven activity. **Workflow (quiz):** topic → lock → practice → question → answer → feedback → next. **Workflow (exam):** click → lock → rules → scrollable questions → answer → submit → result → Review Answer.

ID	P	Requirement	Acceptance
FR-D01	P0	Quiz 1, Quiz 2 and Quiz 3 have independent sets and progress	Tests tab keeps all three
FR-D02	P0	Question types: Multiple Choice, True / False, Short Answer, Blank Space	Each renders and grades
FR-D03	P0	Immediate feedback and explanation in quizzes	Shown before Next
FR-D04	P0	Answers written to local storage within 10 ms	Survives kill and airplane mode

ID	P	Requirement	Acceptance
FR-D05	P0	Mid, Final and Mock lock check before start	Locked exam cannot start
FR-D06	P0	Duration and question count come from the exam set	Rules screen shows Admin values
FR-D07	P0	Exam questions are a scrollable list with a numbered palette and a bottom Next button	Selections preserved when scrolling or jumping
FR-D08	P0	Monotonic timer, auto-submit at zero	Clock rollback cannot extend time
FR-D09	P0	Idempotent submission; late answers rejected	No duplicate final attempt
FR-D10	P0	A **Review Answer** button appears after every exam, Mid, Final, Mock and COC, and after quizzes	Button opens the review list
FR-D11	P0	Review lists every taken question with the student's answer, the correct answer and the description	Order matches the exam
FR-D12	P0	Review is available only after submission	Not reachable during an attempt
FR-D13	P0	Each answer description has Show and Hide buttons, hidden by default, in COC and all exams	State toggles per question
FR-D14	P0	COC Exam is a standalone Home module, absent from the Exams tab	Exams shows only Mid, Final, Mock
FR-D15	P0	COC questions come from one pool and sets may mix universities, years and programs	Set built from many sources
FR-D16	P0	COC practice mode (untimed, feedback) and exam mode (timed)	Mode chosen per paper
FR-D17	P0	COC access: full pool for any active paid entitlement, sample set for others (Admin-configurable)	Rule changes without an app update

ID	P	Requirement	Acceptance
FR-D18	P1	Results shown per Admin policy and linked to attempt and version	Policy applied
FR-D19	P0	Exam integrity metadata and invalid-state rejection	Audit trail exists
FR-D20	P0	Screen-capture blocking applies to quizzes, exams and Review Answer	Same test as FR-C10

8.E Partition E — Commerce, payment and entitlement

Definition: how a student pays and how access is granted, timed and removed. **Workflow:** Section 4.7. Rules PR-1 to PR-7 apply.

ID	P	Requirement	Acceptance
FR-E01	P0	Store lists products, promotional courses and access state	State visible per item
FR-E02	P0	Bundle builder: 7–10 courses of one semester and stream for the bundle price, preselected from My Course List	Submit disabled outside range; server revalidates
FR-E03	P0	Extra courses beyond the maximum use the tiered price (default 100, then 50 ETB)	Summary itemises extras
FR-E04	P0	Single-course purchase before any bundle	Entitlement covers that course only
FR-E05	P0	Full Freshman Package per stream (default 350 ETB); both streams in one order	Unlocks all stream and Common courses
FR-E06	P0	Semester bundle price of 200 ETB is Admin-configurable	Change applies to new orders
FR-E07	P0	Server-side price calculation and stored quote	Client amount is ignored

ID	P	Requirement	Acceptance
FR-E08	P0	Payment instructions for CBE Mobile Banking, CBE Birr, Telebirr with copyable account number	Exact amount shown
FR-E09	P0	Screenshot import, client compression under 300 KB	Oversize returns 422 `RECEIPT_TOO_LARGE`
FR-E10	P0	Transaction reference mandatory and unique	Duplicate returns 409 `TRANSACTION_REFERENCE_DUPLICATE`
FR-E11	P0	Admin verification queue; approval activates entitlements for the exact paid scope	Student opens included courses
FR-E12	P0	Admin-managed duration per product	New entitlements use current duration
FR-E13	P0	Automatic expiry lock	Expired access returns 403 `ENTITLEMENT_EXPIRED`
FR-E14	P0	Students with course-specific access cannot open unpaid courses	Per-course entitlement rows
FR-E15	P1	Renewal extends from `expires_at` when still active; renewal window default 7 days	Alert appears inside the window
FR-E16	P1	Rejected proof can be corrected and resubmitted	Original reason retained
FR-E17	P1	Package supersedes owned single or bundle courses	No double-charge prompt
FR-E18	P0	Order and receipt history for the student	Statuses and reasons visible
FR-E19	P0	Entitlement lifecycle audit	Actor, time, scope recorded

8.F Partition F — Online-first use and synchronization

Definition: the app works online first, keeps local work safe, and syncs once an account exists. **Workflow:** action → local write → outbox row (UUIDv7) → connectivity → sync after phone registration → acknowledgement → mark complete.

ID	P	Requirement	Acceptance
FR-F01	P0	Outbox of unsynced changes survives restart	Verified by restart test
FR-F02	P0	Outbox flushes only after phone registration	No sync calls for local-only users
FR-F03	P0	Idempotent sync endpoint with UUIDv7 operation IDs	Duplicates return `already_applied`
FR-F04	P1	Deterministic conflict resolution	Matrix in Section 12.5
FR-F05	P0	No answer is lost before server acknowledgement	Local state kept
FR-F06	P1	Honest sync status in Learning Status	Never claims synced while pending
FR-F07	P0	Offline grace for downloaded paid items, bounded by an Admin limit	Clock rollback detected
FR-F08	P0	Learning Status screen	Progress, completion, downloads and sync clear

8.G Partition G — Admin operations and Question Bank

Definition: the single privileged role controls every asset. **Workflow:** Section 7.

ID	P	Requirement	Acceptance
FR-G01	P0	Dashboard: students, pending payments, content, expiries	Role-protected
FR-G02	P0	Sidebar item **Questions** with sub-items Mid Exams, Final Exams, Mock Exams, COC Exams	Names and nesting match 7.4

ID	P	Requirement	Acceptance
FR-G03	P0	Question list columns start with Question Number, then University	Order matches 7.4
FR-G04	P0	Filters for University, Course, Year, Stream, Program beside Status	Filters combine
FR-G05	P0	Remove Exam Set Id and Grouped Question Ids from question lists and forms; rename Group Status to Status	Absent from UI and API
FR-G06	P0	Mid and Final form fields per 7.4	Field order tested
FR-G07	P0	COC form fields per 7.4, with Course optional	Field order tested
FR-G08	P0	Mock form fields per 7.4, with no University or Year	Field order tested
FR-G09	P0	Question forms contain no Set Exam Id, Topic, Permission / Access or Time Allowed	Absent
FR-G10	P0	Answer Description field on every question form	Saved and shown in Review Answer
FR-G11	P0	Auto-increment Question Number	Unique and never reused
FR-G12	P0	Dedicated form for each question type	Four forms render
FR-G13	P0	Add Question keeps scope fields for the next entry	Fast bulk entry
FR-G14	P0	Exam set builder with Admin-set duration	Duration reaches the student rules screen
FR-G15	P0	Courses page with free-trial count (FR-B05 to B07)	Saved and audited
FR-G16	P0	Content page per 7.2, with Materials and Tests buttons	Matches 7.2

ID	P	Requirement	Acceptance
FR-G17	P0	Scan to PDF serves COC, Exams, Materials and Tests	Required details per 7.3
FR-G18	P0	Convert 1 to N images, 6 or more fully supported, into one PDF in Admin order	Page count equals image count
FR-G19	P0	Crop, rotate, delete, replace page; non-destructive until convert	Sources retained on failure
FR-G20	P0	Draft, review and publish states with versioning	Published PDF inherits protection
FR-G21	P0	Payment review with duplicate-reference and amount check	Approve or reject audited
FR-G22	P0	Manage products, prices, bundle range, extra tiers, durations, promotions	Rules enforced server-side
FR-G23	P0	Student search, suspend, device revoke, grant or extend access	Reason required; audited
FR-G24	P1	Audit log viewer and reports, including trial usage and conversion	Sensitive actions traceable
FR-G25	P0	Pagination, Delete and Active / Inactive on every tab	Applies to all lists

Added after the design review:

ID	P	Requirement	Acceptance
FR-G26	P0	One shared admin shell (sidebar, top bar, page header, bottom action bar) on every page, per 7.0	Visual review passes on Content, Quizzes, Scan to PDF and Exam Management
FR-G27	P0	Exam sets list per Questions child with a Build set dialog, fields per 7.5, including Exam Set ID, Access / Permission, Year and Allowed Time (min)	Set saves as a draft; duration reaches the student rules screen

ID	P	Requirement	Acceptance
FR-G28	P0	Question form grid per 7.4: Course optional on COC; Mock has no University or Year; formula hint and preview on every question	Layout matches the mockups
FR-G29	P0	Question ID field suggests the next unique ID; Question Number is the auto-incremented identity shown in lists	Duplicate Question ID is rejected
FR-G30	P0	Bottom action bar Reset, Save Draft, Publish with Add Question or Add New Chapter on every authoring form	Same bar on question, quiz and content forms
FR-G31	P0	Test Management under Content → Tests: tabs Quiz 1–3, filters Course, Type, Status, Chapter, columns ID, Question, Course, Type, Chapter, Status, Edit	Lists match the filters
FR-G32	P0	Two-pane content and quiz editor: Course dropdown, Chapters list, Add New Chapter, Free trial checkbox	Free-flagged chapter opens for unpaid students
FR-G33	P0	Scan to PDF wizard in four steps with Category-driven PDF details and no Language field	Required fields per 7.3
FR-G34	P1	Preview as student button on every authoring page	Opens the student view of the draft
FR-G35	P1	Admin light and dark theme toggle in the Admin menu	Contrast 4.5:1 in both themes

8.H Partition H — Notifications and support

ID	P	Requirement	Acceptance
FR-H01	P1	Receipt submitted, approved and rejected notices	In-app, push when available
FR-H02	P1	Expiring and expired notices at Admin thresholds	Alert opens the relevant action
FR-H03	P1	Payment Alerts card on Home	Shows pending, rejected, expiring, expired
FR-H04	P1	Support request from the app, Admin reply	Ticket visible both sides
FR-H05	P2	Content announcements targeted by course	Admin targets

8.I Partition I — Profile, My Course List and Course screen

ID	P	Requirement	Acceptance
FR-I01	P0	After account creation, Profile opens in setup mode	Home reachable only after required fields
FR-I02	P0	Required fields university, program, semester, stream; Save disabled until set	Verified
FR-I03	P0	Cascading pickers; course picker with search, stream chip, price or lock badge, Common always included	Filters correct
FR-I04	P0	Add and remove courses; removal keeps progress and entitlements	Paid-course warning shown
FR-I05	P0	Course screen shows only My Course List, with an empty state and Add courses	No unrelated course
FR-I06	P0	Course screen has a search field on top and filter tabs All, Ongoing, Completed	Tabs filter by progress

ID	P	Requirement	Acceptance
FR-I07	P1	Scope change confirmation; out-of-scope courses leave, progress archived	Verified
FR-I08	P0	Course list stored locally, synced to `student_course` after registration	Offline edits reach the server once
FR-I09	P1	Edit name, phone or email, photo, language, theme	Saved and audited

8.J Partition J — Home, theme and visual system

ID	P	Requirement	Acceptance
FR-J01	P0	Home is card-based with Courses, COC Exam, Learning Status, Payment Alerts, Store Status	Exactly these five cards
FR-J02	P0	No Continue Learning anywhere in the app	UI and API audit finds none
FR-J03	P0	COC Exam card uses the course-card layout	Visual review passes
FR-J04	P0	Light and dark theme toggle on every screen	All screens pass contrast of 4.5:1
FR-J05	P0	Brand and primary colours unchanged	Matches earlier design
FR-J06	P1	Distinct colourful icon per feature and per course	Palette in 6.11
FR-J07	P0	Bottom navigation Home, Course, Progress, Profile	Course active on course screens

9. Non-functional requirements

ID	Area	Requirement	Acceptance
NFR-01	App size	Release package under **20 MB** (arm64 download size from the Android App Bundle)	CI reports size and fails the build at 20 MB or more
NFR-02	Memory	Under 120 MB at runtime on 2–3 GB RAM phones	Profiled on budget reference devices
NFR-03	Local write	Answer persistence under 10 ms at p95	Measured on a budget phone
NFR-04	Online-first	All content loads online first; downloaded items work offline	Airplane-mode test on downloaded items only
NFR-05	Security	TLS, secure token storage, RBAC, device lock, audit	Security review passes
NFR-06	Localization	App interface in English and Amharic across all critical flows	No untranslated critical string
NFR-07	Math and fonts	Bilingual LaTeX and Ethiopic text render with bundled subset fonts	No network request for fonts
NFR-08	Accessibility	Readable type, 48 dp tap targets, labels, 4.5:1 contrast in both themes	Checklist passes
NFR-09	Storage	Large assets served from object storage, never bundled	Bundle audit
NFR-10	Observability	Errors, sync lag, payments, locks and conversion failures measurable	Dashboards and alerts exist
NFR-11	Data cost	Receipt under 300 KB; chapter downloads resumable	Verified on a throttled network
NFR-12	Availability	Admin and API 99.5% monthly; daily backups	Restore drill passes

ID	Area	Requirement	Acceptance
NFR-13	Theming	Light and dark on every screen without restart	Theme switch test

Size governance: Expo production build with Hermes, Android App Bundle with per-ABI splits, no bundled textbooks, PDFs, exam papers or large images, subset fonts only (required KaTeX and Noto Sans Ethiopic weights), vector icons instead of bitmaps, a dependency size review on every pull request, and lazy loading of heavy screens.

10. Technical architecture

10.1 System overview

Layer	Technology	Responsibility
Student app	React Native with Expo, TypeScript, encrypted SQLite, NetInfo	Local account, UI, downloads, outbox, timer, math renderer
Admin portal	Next.js (App Router), TypeScript	Control plane, Question Bank, scan-to-PDF, payment queue
API	NestJS (REST, JWT, validation, rate limiting)	Business rules, entitlements, sync, audit
Database	**PostgreSQL**, UTF-8	Authoritative relational state
Object storage	S3-compatible bucket	Receipts, PDFs, chapter assets, signed URLs
Cache and queue	Optional Redis	Rate limits, jobs for PDF and notifications
Push	Expo push service	Notifications
Tooling	**npm workspaces** monorepo (`apps/mobile`, `apps/admin`, `apps/api`, `packages/shared`)	One lockfile: `package-lock.json`

The server is the authority for device binding, entitlement, price and final exam acceptance. The client is authoritative only for pending local answers until acknowledged.

10.2 Mobile modules

app (boot, navigation, theme) · account (local account, phone registration) · device · academic · home · courses · material (reader) · tests · exams (timer, palette, review) · coc · store · profile · downloads · protection (capture blocking, watermark) · local (SQLite, outbox) · remote (API client) · shared (design system, i18n, icon palette)

10.3 Backend modules

AuthModule · DeviceModule · AcademicModule · ContentModule · DocumentModule (image-to-PDF jobs) · QuestionBankModule · ExamSetModule · CocModule · CatalogPricingModule · OrderModule · PaymentModule · EntitlementModule (expiry scheduler) · SyncModule · DownloadModule (signed, expiring URLs) · NotificationModule · SupportModule · AdminModule · AuditModule

10.4 PDF conversion service

Admin uploads images through signed URLs. A job reads them in Admin order, normalises orientation and size, writes one PDF, computes checksum and page count, stores it and marks the job SUCCEEDED or FAILED. Source images stay until the PDF is published.

10.5 Expiry enforcement

Every protected request checks `entitlement.expires_at` against server time. A scheduled job also sets status EXPIRED and queues notifications. Signed asset URLs last minutes, and downloaded content carries a license expiry that the app checks on open.

11. Security, privacy and content protection

11.1 Transport and sessions

TLS everywhere; tokens in secure storage; short access tokens with refresh rotation; rate limits on login, registration, payment and sync routes.

11.2 Device

Device ID derives from Keystore or Keychain and is stored hashed on the server. Rooted-device and emulator signals are logged for Admin review. Binding starts at first purchase (D4).

11.3 Data

Encrypted SQLite with a key held in secure storage and, once bound, tied to the device and session. Receipts are personal financial data: Admin-only, retained per policy, never logged. Backups are encrypted and restores tested quarterly. Sensitive actions are audited: approvals, price and duration changes, device revocation, publishing, grants.

11.4 Content protection (D9)

Layer	Requirement
Screenshots and recording	Android: `FLAG_SECURE` on all content, exam, quiz, COC and Review Answer screens. iOS: detect capture and hide content, blank on screen recording
Storage	Downloads sit in app-private encrypted storage; no export or share action
Delivery	Signed, short-lived URLs; PDFs rendered in an in-app viewer with no save or print
Watermark	Faint moving overlay with the student's phone or install ID
Accounts	One bound device; Admin-visible alerts on repeated capture attempts

Limits, stated plainly: these controls stop screenshots and screen recording by the operating system. **No app can stop another camera from photographing the screen.** The watermark is the deterrent and the audit trail for that case. Rooted phones can bypass `FLAG_SECURE`, so rooted devices are flagged for review.

12. Data, API and sync

12.1 Headers

Protected requests send `Authorization: Bearer <token>` and `x-device-id: <hardware_uuid>`. Free-content requests send an anonymous install ID instead and receive only free-flagged items. Admin requests use cookie or bearer auth and never send a device header.

12.2 Error codes

HTTP	Code	Meaning
401	UNAUTHORIZED	Missing or invalid token
403	DEVICE_MISMATCH_LOCKED	Account bound to another phone
403	DEVICE_REVOKED	Binding revoked
403	ENTITLEMENT_REQUIRED	No active access
403	ENTITLEMENT_EXPIRED	Duration ended
403	CONTENT_LOCKED	Item outside the free-trial count
409	TRANSACTION_REFERENCE_DUPLICATE	Reference already used
409	PHONE_ALREADY_REGISTERED	Phone belongs to another account
409	SYNC_CONFLICT	Mutation conflict
422	RECEIPT_TOO_LARGE	Image over 300 KB
422	BUNDLE_COUNT_INVALID	Courses outside min and max
422	PDF_CONVERSION_FAILED	Conversion failed
429	RATE_LIMITED	Too many requests
503	SERVER_UNAVAILABLE	Retryable

12.3 Key endpoints

Student: `POST /auth/phone/register` (first purchase) · `POST /auth/login` · `POST /device/bind` · `POST /device/release` · `GET /universities` · `GET /programs?universityId=` · `GET /semesters?programId=` · `GET /courses?semesterId=&stream=` · `GET /courses/picker?semesterId=&stream=&q=` · `GET /courses/:id/material` · `GET /topics/:id/content` · `POST /downloads/token` · `GET /quizzes?courseId=&quizNo=` · `GET /exams?courseId=&kind=&universityId=` · `POST /attempts/:id/submit` · `GET /attempts/:id/review` · `GET /coc/sets` · `POST /orders/quote` · `POST /orders` · `POST /orders/:id/proof` · `GET /entitlements` · `POST /sync/batch`.

Admin: `/admin/questions?kind=&universityId=&courseId=&year=&stream=&programId=&status=` · `/admin/exam-sets` · `/admin/courses` · `/admin/content` · `/admin/scan-jobs` · `/admin/payments` · `/admin/products` · `/admin/devices` · `/admin/audit`. Every admin list supports `page`, `pageSize`, delete and an `isActive` update.

12.4 Sync protocol

The client sends batches of `{operation_id (UUIDv7), type, payload, client_seq}` after phone registration. The server (1) validates token and device, (2) confirms the device owns the account, (3) deduplicates by `operation_id`, (4) validates entitlement, exam ownership and policy, (5) persists and audits, and (6) returns accepted, already_applied, rejected, conflicted or superseded per operation. The client deletes an outbox row only after acknowledgement. The first batch carries the local profile, course list and progress.

12.5 Conflict matrix

Data	Rule
Practice answer	Latest client timestamp wins within an unsubmitted attempt
Exam answer	Accepted only before the server deadline; late answers rejected and logged
Reading progress	Furthest position wins
Final exam submit	First accepted submission wins; later ones return already_applied
Local profile vs server profile at registration	Local values win once, then the server is authoritative
Entitlement	Server only; client changes are ignored

12.6 Data model (PostgreSQL)

Integrity rules: money-affecting writes run in one transaction (order, proof status, entitlements). Timestamps are `timestamptz` stored in UTC. Statuses and types use PostgreSQL enums. The reference check uses a unique index on

`lower(btrim(transaction_reference))`. Entitlements exist per course, so a Full package writes one row per covered course with the same expiry. Every admin-managed table has `is_active boolean` and supports soft delete.

Entity	Key columns and constraints
student	Exists only after phone registration: id, phone UNIQUE, password_hash, status, university_id, program_id, semester_id, stream, profile_completed_at
device_binding	student_id UNIQUE, device_id_hash, status (ACTIVE, RELEASED, REVOKED), bound_at, released_at
university	name, short_name, logo_key, sort_order, is_active
program, semester	Foreign-key hierarchy, sort_order, is_active
course	semester_id, university_id **nullable (NULL = All universities)**, code, title, stream, icon, color, **free_trial_items int**, trial_expiry_days nullable, is_promotional, is_active
module, topic	Course and module parents, sort_order, is_active
content	topic_id, kind (MODULE, NOTE, AMHARIC_NOTE), version, status, title, description, body or asset, answer fields for quizzes, published_at. No language toggle column
document_asset	storage_key, mime, checksum, size_bytes, page_count, source_job_id, status
scan_job, scan_page	job_id, page_no UNIQUE per job, image_key, rotation, crop, target (COC, EXAM, MATERIAL, TEST)

Entity	Key columns and constraints
question	question_number **identity (auto-increment)**, kind (MID, FINAL, MOCK, COC), university_id (null for MOCK), program_id, semester_id, stream, course_id (null for COC), year (null for MOCK), question_type (MCQ, TRUE_FALSE, SHORT_ANSWER, BLANK), body, options jsonb, correct_answer jsonb, **answer_description**, description_asset_key, status (DRAFT, PUBLISHED, INACTIVE), is_free_trial. No set id, topic, permission or time columns
exam_set, exam_set_question	kind (QUIZ1, QUIZ2, QUIZ3, MID, FINAL, MOCK, COC), scope, **duration_min**, shuffle, feedback_mode, result_policy, sample_question_count, version
attempt, attempt_answer	student_id, exam_set_id, state, elapsed_ms, server_started_at; answer UNIQUE(attempt_id, question_id)
product, price_rule	type, price_etb, duration_days, min_courses, max_courses, stream_scope; tiered extra price
orders, order_item	student_id, product_id, quoted_amount, stream, status; UNIQUE(order_id, course_id)
payment_proof	order_id, method, image_key, transaction_reference, status, reviewed_by, reject_reason
entitlement	student_id, course_id, order_id, starts_at, expires_at, status; index on (student_id, course_id, expires_at)
student_course	student_id, course_id, added_at, removed_at, UNIQUE(student_id, course_id). Separate from entitlement
download_license	student_id, content_id, issued_at, expires_at, revoked_at
outbox_event	operation_id UUIDv7 UNIQUE, student_id, type, payload jsonb, applied_at

Entity	Key columns and constraints
notification, support_ticket, audit_log	Actor, type, target, metadata jsonb, timestamps

13. Quality assurance and release gates

Gate	Pass condition
Account	Create account in airplane mode, finish Profile Setup, open free content; database holds no student row
Registration	Phone is requested at first purchase only; local profile and progress then appear on the server
Device	Second phone returns 403 and shows the localized modal; Release then re-bind works
Home	Exactly five cards; no Continue Learning; COC card matches course-card layout
Course	Search plus All, Ongoing, Completed tabs filter correctly
Tabs	Material (Modules, Notes, Amharic Note), Tests (Quiz 1–3), Exams (Mid, Final, Mock only); COC from Home
Reader	Scroll plus bottom Next on all content; no in-page language toggle
Downloads	Download then open offline; delete frees space; expiry makes paid downloads unreadable
Protection	Screenshot and screen recording blocked on content, quiz, exam, COC and review screens
Exam	Scrollable questions; duration from exam set; rollback of system clock does not extend time
Review	Review Answer after every exam and COC; Show and Hide works per description

Gate	Pass condition
Questions admin	Sidebar nesting, column order, filters, removed fields, renamed Status, Answer Description present
Courses admin	All university option, logo, free-trial count, pagination, Delete and Active toggle
Commerce	Bundle 6 and 11 rejected, 7 and 10 accepted; extras priced; Full 350; stream filter correct
Payment	Receipt under 300 KB; duplicate reference returns 409; approve and reject paths
Expiry	Entitlement expires and content locks without an app update; renewal restores it
Admin PDF	6 or more images produce one PDF in Admin order for COC, Exams, Materials and Tests
Theme	Light and dark on every screen
Size	Package under 20 MB; peak memory under 120 MB on a 2 GB device
Out of scope	No GPA and no Continue Learning screen or API exists

Test pyramid: unit (pricing, entitlement, timer maths, free-trial count), API integration on PostgreSQL, mobile component tests, end-to-end on real budget Android phones, security tests (device bypass, forged role, price tampering, forged free-item request), and load tests for the payment queue and sync.

14. Risks, edge cases and mitigations

ID	Risk or case	Required behaviour or mitigation
R1	20 MB is hard with Expo, Hermes and encrypted SQLite	Measure a bare Expo build in Phase 1 before committing; use App Bundle arm64 splits, subset fonts, lazy modules; escalate early if the gate cannot be met

ID	Risk or case	Required behaviour or mitigation
R2	Free content has no account, so it can be shared	Serve free items as rate-limited, watermarked, public-flagged assets; keep free counts small
R3	Local data lost on uninstall before first purchase	Warn the student (FR-A13); offer no recovery because no remote copy exists
R4	One person creates many local accounts to repeat free trials	Accept; free portion is limited; Admin may later add per-install limits
R5	Photo of the screen with another camera	Cannot be blocked; watermark and audit (11.4)
R6	Student owns single course then buys Full package	Package covers it; no second charge
R7	Entitlement expires during an exam	Attempt may finish; no new exam starts (Admin policy flag)
R8	Phone lost or replaced	Release from the old phone, or Admin revokes after identity check
R9	Clock changed	Monotonic timer governs exams; server time governs expiry
R10	Receipt reused, fake or wrong amount	Unique reference; Admin checks the bank record; reject with reason, or approve partial scope
R11	PDF conversion failure	Retryable; sources retained
R12	Low storage	Warn before downloads; offer delete of least-recent downloads
R13	App killed mid-quiz	Answers restored from local storage
R14	Admin deletes a course with student progress	Soft delete; prefer Inactive; progress archived

15. Traceability and definition of done

Partition	IDs	Mandatory acceptance
A Identity and device	FR-A01 to A13	Local account, deferred registration, one-device lock
B Catalog	FR-B01 to B10	Scope, stream, All university, free-trial count
C Content	FR-C01 to C17	Reader, downloads, protection
D Assessment	FR-D01 to D20	Quizzes, exams, COC, review
E Commerce	FR-E01 to E19	Pricing, proof, entitlement, expiry
F Sync	FR-F01 to F08	Outbox, honest sync
G Admin	FR-G01 to G25	Question Bank, courses, content, scan
H Notifications	FR-H01 to H05	Alerts, support
I Profile	FR-I01 to I09	Setup, course list, Course screen
J Home and theme	FR-J01 to J07	Cards, themes, icons

A requirement is done when it is implemented, tested, authorized on the server, has happy-path and failure UI, has database constraints where money or access is involved, has defined offline behaviour, writes audit records for sensitive actions, and ships inside the 20 MB package.

Removed or changed since v2.0: Continue Learning (FR-044), one-bilingual-record Amharic (FR-023), Access and Topic on questions, per-course COC breakdown (FR-185), device registration at launch, offline-first caching, 50 MB, MySQL, pnpm.

16. Implementation roadmap

Priority per the 12 October instruction: review both codebases, align with the new structure, work in small sub-phases, and deliver the **Question Bank first**.

Phase	Scope	Exit criteria
0	Review Admin and Student codebases; align folders, designs and this SRS; close open questions	Approved baseline and gap list
1	npm workspaces monorepo, PostgreSQL migrations, CI with 20 MB size gate, bare-build size measurement	Pipeline green; size feasibility confirmed
2a	Question table, enums, identity Question Number, API with filters and pagination	FR-G03 to G05, G11 pass
2b	Admin Questions sidebar and nested lists (Mid, Final, Mock, COC), filters, Status, Delete and Active toggle	FR-G02 to G04, G25 pass
2c	Question forms per kind and per question type, Add Question, Answer Description, Publish	FR-G06 to G10, G12, G13 pass
2d	Exam set builder with Admin duration	FR-G14 passes
2e	Student exam screens: scrollable list, palette, Review Answer, Show and Hide	FR-D07, D10 to D13 pass
3	Admin Courses page (All, logo, free-trial count) and catalog	FR-B01 to B09 pass
4	Admin Content page, Scan to PDF for COC, Exams, Materials, Tests	FR-G16 to G20 pass
5	Student app shell: local account, Profile Setup, card Home, Course screen, themes	FR-A01 to A02, FR-I, FR-J pass
6	Reader, quizzes, COC module, free-trial enforcement	FR-C, FR-D pass
7	Phone registration, device binding, commerce, receipts, entitlements, expiry	FR-A03 to A13, FR-E pass
8	Downloads, content protection, sync	FR-C07 to C11, FR-F pass
9	Notifications, support, reports, hardening, pilot at one university	All release gates pass

17. Open questions for the product owner

#	Question	Assumption used
Q1	Should device binding start at first purchase (as written) or at account creation?	At first purchase, because the account is local until then
Q2	Is exam duration set on the exam set (as written), or should a field exist elsewhere?	On the exam set only
Q3	Where are Quiz 1–3 questions authored now that the Quiz tabs are gone from Questions?	In Admin Content → Tests (confirmed by the Test Management screens); students keep Quiz 1–3
Q4	The COC form design shows an optional Course field. Should COC results show a per-course breakdown?	Yes, for questions tagged with a course
Q5	Does 20 MB mean the Play Store download size (arm64) or the universal APK?	Arm64 download size
Q6	How is the phone number verified at first purchase (SMS code, or only collected)?	SMS code, with a manual Admin fallback
Q7	Is the Question Number unique across all kinds, or restarted for each of Mid, Final, Mock and COC?	One identity per kind
Q8	May students see other universities' papers, or only their own?	All universities, own first
Q9	Free-trial count: do quiz chapters, exam samples and COC samples have their own counts, or one number per course?	One number for modules, notes and quiz chapters; exam and COC samples configured separately
Q10	Default duration per product	One semester (about 150 days) for bundles, one academic year for Full

18. Appendices

Appendix A — Student flow map

Install → Splash → Create local account → Profile Setup → Home. Home → Courses → Course (search, All / Ongoing / Completed) → Material (Modules, Notes, Amharic Note → lock → scroll, Next) · Tests (Quiz 1–3 → lock → practice → feedback → Review Answer) · Exams (Mid, Final, Mock → lock → scrollable questions → submit → Review Answer). Home → COC Exam → lock → practice or timed → submit → Review Answer. Home → Learning Status · Payment Alerts · Store Status → Store → phone registration → device bind → pay.

Appendix B — Payment and expiry lifecycle

Quote → register phone (once) → pay → screenshot → compress → reference → pending → (reject → reason → resubmit) or (approve → entitlements → active → expiring → expired → locked → renew).

Appendix C — Glossary

Term	Definition
Local account	Account stored only on the phone until the first purchase
Entitlement	Server-granted access to one course for a period
Free-trial items	Number of items per course open without payment
Bundle	7–10 courses of one semester and stream for one price
Full Freshman Package	All courses of a stream for the freshman year
COC Exam	Cross-course module opened from Home, separate from Exams
Review Answer	Post-exam list of questions, correct answers and descriptions
Outbox	Local queue of unsynchronised changes
Monotonic clock	Elapsed-time source unaffected by wall-clock changes

Appendix D — Final acceptance checklist

- Package under 20 MB; memory under 120 MB on a 2 GB phone
- Stack is Expo, Next.js, NestJS, PostgreSQL, npm
- Local account works offline; phone registration happens only at first purchase
- Home has Courses, COC Exam, Learning Status, Payment Alerts, Store Status; no Continue Learning
- Course screen has search and All, Ongoing, Completed tabs
- Material tabs Modules, Notes, Amharic Note; no language toggle inside content
- Scroll-first reading with a bottom Next button on all content
- Online-first delivery with download and delete
- Screenshot and recording blocked on all content screens
- Review Answer with Show and Hide after every exam and COC
- Questions sidebar, list, filters and forms exactly as in 7.4
- Courses page with All university, logo, free-trial count, pagination, Delete and Active toggle
- Scan to PDF for COC, Exams, Materials and Tests; 6 or more images to one PDF in Admin order
- One-device lock and release; bundle 7–10; extras; single course; 350 ETB Full package
- Manual receipt flow with unique reference; automatic expiry
- Light and dark theme with colourful icons; no GPA